Review Note: EuroSAT Land-Cover Classification Project

I have gone through the code guide, the classification report and train_history.csv. The baseline is a good start and the test result of about 99% is encouraging. But the work is not yet ready to go into a paper. Please read the points below, fix them in the order given, and send me the results before moving to the next step.

1. What is wrong in the current work

1. The training stopped at epoch 16, and the best validation loss is also at epoch 16. Early stopping did not trigger, so the model had not finished training. Tell me why the run stopped there.
2. Only one run with one seed was done. One run cannot show whether the result is stable.
3. The model uses BatchNorm. BatchNorm does not work well when data is imbalanced or split across clients, and we will need that later. We must compare it with GroupNorm.
4. The 64x64 images were enlarged to 224x224. This makes training about 12 times slower and adds no new information. We need to check if a smaller size gives the same accuracy.
5. The split is random at image level. Neighbouring satellite patches may fall in both train and test, so 99% may be higher than the real performance. This must be mentioned in the paper. We should not compare our number with published results.
6. The code has no client_id option and does not save per-class or per-client results. We need both for the next phase.

2. What to correct

Rules for all runs from now on:

- Use the same fixed splits.json every time. Do not create a new split.
- Choose settings using the validation set only. Use the test set once, at the end.
- Train with up to 100 epochs and stop when validation loss does not improve for 10 epochs. Note the best epoch.
- Run three seeds (42, 123, 2024) and report mean and standard deviation.
- Keep the final paper numbers for five seeds.

Runs to do:

1. Pretrained ResNet-18 with BatchNorm at image sizes 224, 96 and 64.
2. ResNet-18 from scratch with BatchNorm.
3. ResNet-18 from scratch with GroupNorm.

For 2 and 3, use the smaller image size if it loses less than half a percent in macro-F1 compared to 224.

Code changes:

- Add a client_id option in data.py.
- Add a norm option (batch or group) in model.py.
- Add a loss option in train.py (cross-entropy for now).
- Save accuracy, macro-F1 and per-class recall for every run.
- Write one line per run in a single results.csv file, so that all tables in the paper can be made from this

file and no number is typed by hand.

Send me for every run: config.json, splits.json, history.csv, training curves, test metrics, classification report and confusion matrix.

Be ready to explain your code to me, especially how the split is made and why the best model is chosen on validation loss.

3. Next experiments (start only after step 2 is checked)

A. Imbalanced data. Create two kinds of imbalance and train on them. Validation and test sets must stay balanced.

- Dirichlet split into 5 and 10 clients with alpha values 0.1, 0.5, 1 and 10. Use the partition code from NIID-Bench and cite it.
- Long-tailed EuroSAT, where some classes have far fewer images, with imbalance ratios 10, 50 and 100.

B. Imbalance-aware training. Compare plain cross-entropy with weighted cross-entropy, class-balanced loss and focal loss. Choose beta and gamma on validation only. Report macro-F1, balanced accuracy and recall of the rarest classes, not only accuracy.

A model trained on a skewed split will obviously do worse. That alone is not a result. The useful result is how much the new loss recovers.

4. What to write for the paper

Write one section at a time in LaTeX and send it to me before starting the next.

1. Related work. Four short parts: EuroSAT and remote sensing classification, non-IID data in federated learning, class imbalance losses, and federated learning in remote sensing. End with a paragraph saying what our work does that the closest paper does not.
2. Problem definition. The task, the Dirichlet split, the long-tailed setting and the metrics, with proper notation.
3. Method. Model, normalization choice and loss functions with equations.
4. Experimental setup. Dataset table, split, all settings, seeds, hardware and software versions.
5. Results. Tables and figures, each with a short explanation of what it shows.
6. Discussion and limitations. Must include: random image-level split, one dataset only, RGB only, small number of seeds, and that Dirichlet splits only approximate real regional differences.
7. Introduction and abstract. Write these last, when the numbers are final.

Figures needed: training curves, confusion matrix, client-by-class distribution plot, macro-F1 against alpha and against imbalance ratio, per-class recall comparison. Save all as PDF.

Please do not write anything like state-of-the-art, and do not copy numbers from other papers into our tables. Every claim must be supported by our own results.

Send the work phase by phase. Do not move on until I have confirmed the previous one.